



HWA CHONG INSTITUTION (COLLEGE SECTION)
2018 JC2 9744 H2 BIOLOGY
PRELIMINARY EXAMINATION PAPER 2 MARK SCHEME

QUESTION 1

- (a) Name the organelles **A** and **B** shown in Fig. 1.1. [2]

A: Rough endoplasmic reticulum

B: Nucleus

- (b) Explain how organelles **A** and **B** are adapted to allow the goblet cell to secrete mucus out of the cell. [4]

Organelle A

Structure

- Organelle A contains extensive network of cisternae

Function

- Provides increased membrane surface area, for the embedment of ribosomes
- lumen allows folding of polypeptide into native conformation

Organelle B

<i>Structure</i>	<i>Function</i>
• The nuclear envelope	• protects DNA from degradation
• The nuclear envelope is perforated with pore	• to enable movement of mRNA from the nucleus to cytoplasm
• It contains genes	• that codes for proteins

- (c) Describe how the similarities between mitochondria and prokaryotic cells support the theory of endosymbiosis. [2]

- Both mitochondria and prokaryotic cells have ribosomes which are similar, in terms of size/ RNA sequences as compared to cytoplasmic ribosomes of eukaryotic cells
- Like prokaryotic cells, mitochondria have circular DNA which are not associated with histones
- The inner membranes of mitochondria have enzymes that are similar to those found in the plasma membranes of prokaryotic cells

[Total: 8]

QUESTION 2

- (a) Describe what is meant by facilitated diffusion. [1]
- Transport of solutes down a concentration gradient to aid of a transport protein embedded in the membrane
- (b) Identify which transporter is involved in facilitated diffusion in Fig. 2.1. [1]
- Transporter Q
- (c) Explain how glucose is transported from the intestinal lumen to the extracellular fluid. [4]
- Glucose entering the cell together with sodium ions via transporter P
 - By active transport
 - Glucose leaving the cell via the transporter Q
 - By carrier-mediated facilitated diffusion
- (d) Suggest why R is necessary in facilitating the transport of glucose across the epithelial cell in Fig. 2.1. [2]
- Protein R keeps a low concentration of Na^+
 - To drive the glucose transporter P

[Total: 8]

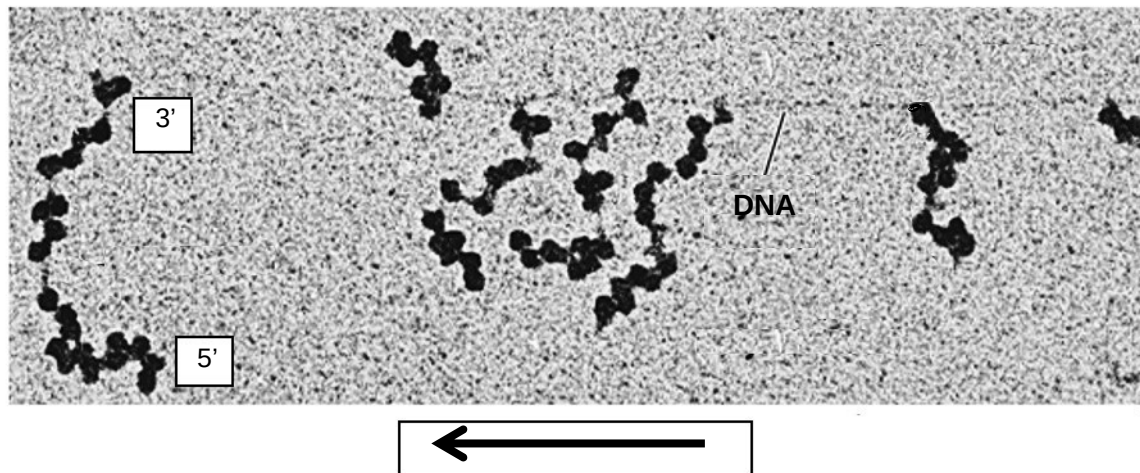
QUESTION 3

- a) Describe the main features of ribosome structure. [2]
- Ref. to the ribosome comprising of both ribosomal proteins and ribosomal RNA (rRNA)
 - The ribosome consists of small and large ribosomal subunits
 - Small ribosomal subunit has a mRNA binding site
- b) State the role of the rS2 promoter. [1]
- The rS2 promoter contains RNA polymerase binding site to initiate formation of transcription initiation complex
- c) List three ways by which process X differs from process Y. [3]
- Point to point comparisons on the following points
- Ref. to different templates
 - Ref. to different enzymes used
 - Ref. to type of covalent bonds formed
 - Ref. to location of cell where the process occurred.
- d) Suggest how a mature ribosome is assembled upon the completion of process Y. [3]
- The synthesized rS2 polypeptide folds into its specific 3D conformation
 - Ref. to where it associates with rRNA
 - Ref. to assembly of small and large ribosomal subunits in cytoplasm to form a mature ribosome

[Total: 9]

QUESTION 4

Fig. 4.1 is an electron micrograph showing ribosomes involved in protein synthesis in an *Escherichia coli* cell.



(a)(i) On Fig. 4.1,

- label the 3' and 5' ends of the polynucleotide in the square boxes provided
- draw an arrow in the rectangular box provided to indicate the direction of transcription. [2]

(ii) The mechanism of protein synthesis shown in Fig. 4.1 is unique to prokaryotes.

Explain why this is so. [4]

- Simultaneous transcription and translation occurs uniquely in prokaryotes
- Due to the absence of nuclear membrane/ envelope
- mRNA can be degraded easily/ unstable
- Thus, simultaneous transcription and translation occurs in order to speed up the process of protein synthesis

(b) In *E. coli* cells, transcription is regulated by a cluster of genes known as an operon, which can be either inducible or repressible. The *trp* operon is one such operon.

(i) Explain if the *trp* operon is inducible or repressible. [2]

- The *Trp* operon is a repressible operon
- The *Trp* operon is normally switched on and switched off upon accumulation of the tryptophan

(ii) Describe what happens when tryptophan is absent in *E. coli*. [3]

- The *trp* repressor acts as a corepressor, which is absent
- Thus, the *trp* repressor is in an inactive conformation and cannot bind to the operator
- RNA polymerase can bind to promoter to allow the transcription of *trp* operon genes/ the operon to be switched on

- (c) *E. coli* was cultured in a growth medium with all the essential nutrients except tryptophan. Fig. 4.2 shows the changes in concentration of enzymes involved in tryptophan synthesis over time.

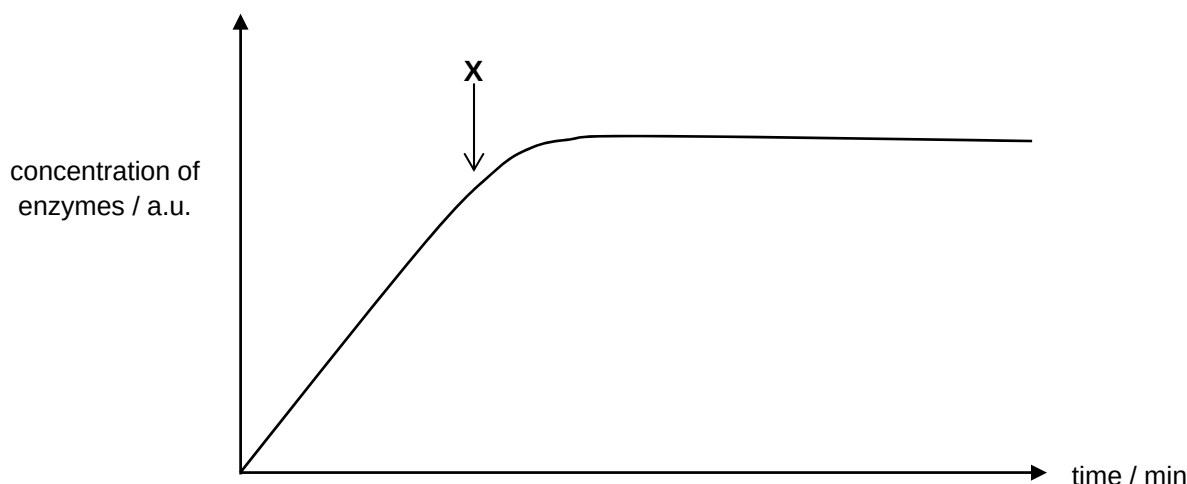


Fig. 4.2

As tryptophan is intrinsically synthesised, it is also utilised by *E. coli* at the same time.

Draw an arrow with the label **X** on Fig. 4.2 to indicate the time at which excess tryptophan was added to the growth medium. [1]

[Total: 12]

QUESTION 5

- (a) Describe the effect of increasing water level on the mean length of the submerged internodes and the mean concentration of ethylene in rice stems. [4]

- As depth of water increases, mean length of internodes increases
- 2 depths + 2 mean lengths
- As depth of water increases to 30 cm, mean concentration of ethylene increases to a maximum of $1.1 \text{ mm}^3 \text{ dm}^{-3}$
- As depth of water increases to 60 cm, mean concentration of ethylene plateaus at $0.9 \text{ mm}^3 \text{ dm}^{-3}$

- (b)(i) Suggest how CEND is activated. [2]

- Active ETR1 causes dephosphorylation of CEND-EIN2 complex
- CEND dissociates from EIN2

- (ii) Explain how CEND increases the rate of transcription of the ethylene response gene. [4]

- Active CEND diffuses into nucleus
- Active CEND binds inactive EIN3 and activates EIN3
- Ref. to active EIN3 binding enhancer of ethylene response gene
- Ref. to stabilisation of transcription initiation complex

- (iii) Flooding is an indirect consequence of global warming. Suggest how scientists can make use of these findings to mitigate the effects of flooding. [1]

- Flooding of crop land can become more frequent / intense
- Ref. to genetic modification of crops to increase flood tolerance
- Ref. to maintenance of a steady food supply

[Total: 11]

QUESTION 6

(a) Identify and describe what happens during the longest stage in a cell cycle. [3]

- Ref to interphase

G₁ phase of interphase

- Cells increase in size and acquire ATP
- Intensive cellular gene expression occurs

S phase of interphase

- Each DNA molecule undergoes semi-conservative DNA replication

G₂ phase of interphase

- Centrioles replicate and the mitotic spindle begins to form

(b) Using the information shown in Fig. 6.1, describe and explain the effect of colchicine on mitosis. [4]

- Untreated cells alternated between interphase and mitosis every 30 minutes whereas colchicine-treated cells entered mitosis at 75 minutes but remained there for hours
- Prevent degradation of cyclins
- Inactivation of cyclin-dependent kinases (Cdks), hence mitosis cannot be completed
- Ref. to disruption of the formation of spindle fibre leading to incompleteness of mitosis

(c) Research has shown that Colchicine exerts its effect on tubulin molecules that make up the microtubules. It is proposed that colchicine could be used to treat cancer.

(i) Explain how colchicine could be used for the treatment of cancer. [2]

- Colchicine inhibits the assembly of microtubules
- Chromosomes are unable to attach to microtubules
- Cell division cannot proceed and hence uncontrolled cell division is prevented

(ii) Suggest a potential side effect of using colchicine in cancer treatment. [1]

- Colchicine is not cell-specific and hence, (even) normal cells cannot undergo cell division

[Total: 10]

QUESTION 7

(a)(i) Using the formula above, calculate the value for χ^2 . Show your working clearly. [2]

$$\frac{(10-10.5)^2}{10.5} + \frac{(4-3.5)^2}{3.5} = 0.0952$$

(ii) Given the critical value for χ^2 at $p = 0.05$ with one degree of freedom is 3.84, draw a conclusion and explain why this conclusion should be treated with caution. [3]

- χ^2 calc (= 0.095) < χ^2 crit (= 3.84)
- probability that observed number differs from expected ratio of 3:1 due to chance > 0.05, not significant
- it is based on a small sample

(b) Using the symbols for the alleles stated on page 23, draw a genetic diagram to explain the results shown in Table 7.1. [5]

Parental phenotypes ash-red female, no crest ash-red male, crest

Parental genotypes $Z^A W D D$ $\times$ $Z^A Z^B d d$

Gametes $W D$ $Z^A D$ $Z^A d$ $Z^B d$

Punnett square

		Male gametes	
		$Z^A d$	$Z^B d$
Female gametes	$W D$	$Z^A W D d$	$Z^B W D d$
	$Z^A D$	$Z^A Z^A D d$	$Z^A Z^B D d$

Offspring genotypic ratio 1 $Z^A W D d$: 1 $Z^B W D d$: 1 $Z^A Z^B D d$: 1 $Z^A Z^A D d$

Offspring phenotypic ratio $\underbrace{1 \text{ ash-red female, no crest} : 1 \text{ blue female, no crest}}_{3} : \underbrace{2 \text{ ash-red male, no crest}}_{7}$

Observed numbers 3 4 7

(c) Suggest a method for the breeder to determine the genotype of a blue male pigeon with crest. [2]

- Cross with brown female
- If 50% brown, 50% blue in males, male is heterozygous at that gene locus

[Total: 12]

QUESTION 8

- (a) For each of these examples, explain the effect on the rate of photosynthesis of one type of plastid changing into another type of plastid. [4]

Example 1

- Rate of photosynthesis increases
- Ref to explanation on how the rate of photosynthesis is affected

Example 2

- Rate of photosynthesis decreases
- Ref. to explanation on how the rate of photosynthesis is affected

- (b)(i) Define the term *absorption spectrum* for photosynthesis. [1]

A graph of the amount of light absorbed at different wavelengths by the cyanobacteria / photosynthetic pigment

- (ii) With reference to Fig. 8.1, explain the effect of different lighting conditions on the absorption spectra of the two groups of cyanobacteria. [3]

- Ref. to data about cyanobacteria grown in fluorescent light vs in red light
- Ref. to reasons to support the effect of different lighting conditions on the absorption spectra of the cyanobacteria

[Total: 8]

QUESTION 9

- (a) Suggest how the ability to use echolocation may have evolved from an ancestor that did not have that ability. [4]

- Selection pressure is lack of competition for food / prey
- Individuals with allele(s) / gene(s) for echolocation have a selective advantage
- Echolocation allele(s) / gene(s) passed on to next generation
- Over many generations frequency of echolocation allele increases

- (b)(i) Name the genus to which the *P. pygmaeus* belongs. [1]

Pipistrellus

- (ii) Using the data in Table 9.1, suggest why pipistrelles were originally classified as one species. [2]

- Ref. to similar / same, (body) mass at 5.5g
- Ref. to similar wingspan at 0.21-0.22m
- Ref. to similar / same, colour at medium to dark brown

- (c)(i) Using the information given in Table 9.1 and Fig. 9.1, discuss how both morphological and

molecular data support the findings that *P. pipistrellus* and *P. pygmaeus* belong to different species. [3]

- Morphological data indicate different range of echolocation but both species show structural / anatomical homologies
- Molecular data as in *cyt b* gene sequence homology shows differences in the substitution site
- *P. pygmaeus* and *P. pipistrellus* are in different clades I & II respectively

(ii) In more recent phylogenetic studies using other forms of molecular data, scientists have shown that *P. pipistrellus* and *P. pygmaeus* may have undergone convergent evolution in Europe, the Mediterranean and Asia.

Suggest how this is possible. [2]

- Ref. to evolving independently in different areas / under similar selection pressure, Europe, the Mediterranean and Asia
- Ref. to bats living in similar ecological niches / subjected to similar environments, hollow trees / caves and have limited flying range

[Total: 12]

QUESTION 10

(a) (i) State the location in the body where macrophages have their origin. [1]

Bone marrow

(ii) Describe the mode of action of a macrophage. [3]

- Detection / recognition
- Engulfs / envelops, pathogen / bacterium
- Forms, vacuole / vesicle / phagosome
- Ref. to lysosome fusion
- Ref. to hydrolytic / digestive, enzymes

(iii) It is sometimes possible for *M. tuberculosis* to survive within macrophages.

Suggest **one** way in which *M. tuberculosis* may survive within a macrophage. [1]

- Prevent fusion of phagosome with lysosomes
- Produces inhibitors for / deactivates, lysosomal enzymes or forms resistant spore / is resistant to digestive enzymes

(b) The standard treatment for TB continues for six months and initially involves the use of four different antibiotics. [2]

- Increases chances of, killing, all bacteria
- If bacteria are resistant to one antibiotic, then still susceptible to other antibiotics
- Reduces chance of mutations arising / if bacteria susceptible mutations unlikely to occur against all antibiotics

(c) TB is a disease of global importance.

Discuss the factors influencing the trends shown in Table 10.1.

[3]

- Overall trend for both is increase
- Rate of increase in cases of MDR-TB greater than rate of increase in cases of TB
- Ref. to problems with people living in, poorly ventilated accommodation / close proximity to each other
- Vaccination programmes not able to prevent increase / herd immunity insufficient
- Qualified ref. to link between, HIV / HIV/AIDS, and TB
- Less money available for, longer drug treatment / vaccination, by, health authorities / governments
- Ref. to some success of, prevention / treatment programme

[Total: 10]